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Location Estimation via the Direction of Arrival Techniques Based on the IEEE 802.11n WLANs

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ABSTRACT

Location estimation techniques have evolved recently and became an essential part in many applications. These techniques include received signal strength indication, finger printing, time of arrival, time difference of arrival, and direction of arrival. All the techniques, other than the last technique, are applicable through all the IEEE 802.11 standards. The direction of arrival can only be estimated via access points that implement the IEEE 802.11n, ac standard amendments that included Multiple Input Multiple Output antennas in its air interface. The required time synchronization and computational intensive triangulation for location estimation with the time-based methods are not needed if direction of arrival is used. In this paper, the WLAN standard IEEE 802.11n is tested for the maximum ranges of accurately estimated direction of arrival in indoor environments. The antenna inter-element spacing that maximizes this range is determined. The accuracy of angle estimation vs. signal to noise ratio is calculated. The wireless channel is modelled by the ray tracing model which enables determining all the information about incidence angles for different signals that reach the receive antenna elements. Different direction of arrival estimation algorithms are tested.

Keywords: WLAN, IEEE 802.11n, Positioning, DoA, Super Resolution, MIMO.

CONCLUSION

Location estimation using the direction of arrival (DoA) detection is a promising technique for the IEEE 802.11n, ac, as they introduced the MIMO for the air interface. MIMO-OFDM coding with OSTBC is used for DoA estimation through the IEEE802.11n WLAN signals in indoor environments. DoA estimation algorithms such as beam scan (BS), minimum variance distortion-less response (MVDR), multiple signal classification (MUSIC), and estimation of signal parameters via rotational invariance technique (ESPRIT) are compared regarding the range of accurate and contiguous estimation of DoA and the effect of the inter-element spacing on the estimation accuracy at the receiving antenna. The MIMO Wireless Channel was simulated via the ray-tracing model (RTM) for gaining all the details of the individual signal rays comprising the received multipath signals. It is found that the increase in the number of antenna elements at the transmitter decreases the performance of the estimation techniques as it introduces more interferers, which deteriorate the estimation accuracy. However, MUSIC and ESPRIT techniques proved their accuracy for DoA estimation in indoor environments. It is noted that they can support a wide range of incidence-angles, compared to BS and MVDR, while keeping the maximum accuracy. It is also proved that the antenna elements at the receivers should be spaced by approximately 0.5λ for maximum contiguous measurable DoA angle range if MUSIC or ESPRIT is used for DoA estimation. This is important to keep the system capacity from reduction by the inter-element coupling. Thus, MUSIC and ESPRIT is the most suitable techniques for DoA estimation through WLANs in indoor environments.

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S. A. Napoleon, A. S. Omar, S. H. Elramly, S. A. Khamis, M.E. Nasr, "Location Estimation via the Direction of Arrival Techniques Based on the IEEE 802.11n WLANs," *International Journal of Research in Wireless Systems (IJRWS)*, Vol. 3, Issue No. 1 , pp. 41-48 , April, 2014.

ISSN: 2320 - 3617

I. INTRODUCTION

Indoor location estimation has evolved recently as a complement to the global positioning system (GPS) coverage for outdoors. The location estimation techniques can be classified as: signal strength based, delay based and angle based. Signal strength based methods include received signal strength indication (RSSI) and finger printing (FP). Details for these techniques can be found in [1]. Time based methods include time of arrival (ToA), time difference of arrival (TDoA) techniques. The details for these techniques can be acquired from [2]. Angle based techniques depend on estimating the direction of arrival (DoA) for the received signals from at least two transmitting antenna arrays. Location can be calculated through a simple triangulation. It is noted that triangulation for calculating location through the time based methods is more complex than the triangulation for location calculation by DoA [3]. Also, it is noticeable that signal strength based techniques are site-specific and require the building of a look-up database [3]. In this context, many DoA estimation algorithms are employed for location estimation such as beam scanning (BS), minimum variance distortion-less response (MVDR), and super resolution algorithms such as multiple signal classification (MUSIC) and estimation of signal parameters via rotational invariance techniques (ESPRIT). Details about these estimators are discussed in [3], [4]. Combining the above with the fact that new standards for wireless lan (WLAN), such as IEEE 802.11n, ac, include multiple input multiple output (MIMO) antennas in its air interface, which prompted using the DoA estimation algorithms for location estimation. In these WLANs, the MIMO is combined with orthogonal frequency division multiplexing (OFDM). Space time block code (STBC) is used as well to get the benefits of MIMO, which are increasing communication data rates and overcoming multipath effects. Since the details for multipath signals in terms of incidence angles and path delays, the indoor multipath channel is simulated using a ray tracing model (RTM) [5]. A comparison between the DoA estimation algorithms: BS, MVDR, MUSIC and ESPRIT is performed by calculating estimation error vs. MIMO antenna element spacing normalized to the signal wavelength in order to find ranges of antenna element spacing for least attainable estimation error under these conditions. The paper is organized as follows; section II surveys different direction of arrival estimation techniques, section III shows how the MIMO wireless channel is modelled, section IV describes the ray tracing model (RTM), section V shows the evaluation system setup, section VI is dedicated for results and discussion, and the conclusion can be found in section VII.

II. DIRECTION OF ARRIVAL ESTIMATION ALGORITHMS

To measure the DoA, an antenna array is needed. Such an antenna is illustrated in Fig. 1.

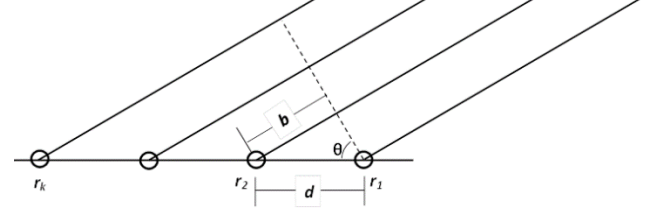


FIG. 1: DOA MEASUREMENT VIA AN ANTENNA ARRAY

where d is the distance separating the antenna elements and it is considered a constant and $b = d\sin(\theta)$. Clearly, the ray arriving r_2 will be delayed in time by $\tau = \frac{d\sin(\theta)}{c}$ seconds compared to r_1 where c is the propagation velocity, or has a phase delay of $\phi = \frac{\omega_c d\sin(\theta)}{c}$ radians where ω_c is the carrier frequency [3][22]. In general, ϕ and τ can be expressed as

$$\begin{aligned}\tau_i &= \frac{(i-1)d\sin(\theta)}{c} \\ \phi_i &= \frac{(i-1)\omega_c d\sin(\theta)}{c} \\ i &= 1, 2, \dots, k\end{aligned}\quad (1)$$

It is assumed also that the receiving antenna array is in the far field [6] i.e.

$$\begin{aligned}l &> \frac{2D_a^2}{\lambda_c} \\ l &\gg \lambda_c \\ l &\gg D_a\end{aligned}\quad (2)$$

where l is the distance between the antenna array and the source, while D_a is the maximum dimension of the array. To remove ambiguity, values of the phase delay, ϕ , should be restricted to $|\phi| < \pi$. and this can be achieved if the element separation, d , is restricted to $d < \frac{\lambda_c}{2}$ [22], $\lambda_c = \frac{2\pi c}{\omega_c}$. The received signal at any element may be written as

$$r_m(t) = H_m(\omega_c, \theta_i) e^{-j(m-1)\phi_m} s_m(t) + n_m(t), \quad m = 1, 2, \dots, k \quad (3)$$

where H_m is the transfer function for the m^{th} element and $n_m(t)$ is the additive white Gaussian noise (AWGN) at that element. In case of omni-directional and identical antenna elements, H_m can be considered constant and can be impeded in the amplitude of $s(t)$. The above equation can be represented in a matrix form as

$$\mathbf{r}(\mathbf{t}) = \mathbf{a}(\theta) \cdot \mathbf{s}(\mathbf{t}) + \mathbf{n}(\mathbf{t}) \quad (4)$$

where $\mathbf{r}(\mathbf{t})$, $\mathbf{a}(\theta)$, $\mathbf{s}(\mathbf{t})$, and $\mathbf{n}(\mathbf{t})$ are the matrix notations for these variables and If all antenna elements assumed to be omni directional and identical (which is the case in MIMO) then

$$\mathbf{a}(\theta) = [1 \ e^{-j\omega_c \tau} \ e^{-j2\omega_c \tau} \ e^{-j(m-1)\omega_c \tau}]^T \quad (5)$$

The vector, $\mathbf{a}(\theta)$, in (5) is the steering vector and T denotes the transpose of it. To estimate the unknown θ , various techniques are summarized below. They may be classified as nonparametric and parametric methods.

A. Beamscan

It is one of the nonparametric techniques. In it, the steering vector is scanned over all the values of θ . The value of θ is the one that maximizes $P(\theta)$ which is given by (6).

$$P(\theta) = \mathbf{a}^H(\theta) \cdot R \cdot \mathbf{a}(\theta) \quad (6)$$

$$R = \frac{1}{N} \sum_{i=1}^N \mathbf{r}(iT_s)^H \cdot \mathbf{r}(iT_s) \quad (7)$$

where R is an estimate of the signal's covariance, $(\cdot)^H$ denotes the Hermitian of the matrix, N is the number of snapshots or samples and T_s is the sampling interval. The algorithm resolution depends on the number of antenna elements. As the number of elements increase, the resolution becomes better [7], [8]. A signal includes multiple correlated sources, as in the case of multipath scenarios, is a common phenomenon in indoor locations due to the multiple reflections from walls and other obstacles which will introduce inconsistency in estimation [8].

B. Minimum Variance Distortion-less Response (MVDR)

This technique is also belongs to the nonparametric techniques. It attempts to minimize the output power of the receiving antenna array while keeping the transfer function of the array at unity i.e.

$$\mathbf{W}^H \cdot \mathbf{a}(\theta) = 1 \quad (8)$$

where $\mathbf{W}$ is a weighting matrix. This will yield the following spectrum which will peak at the value of the DoA.

$$P(\theta) = \frac{1}{\mathbf{a}(\theta)^H \cdot R^{-1} \cdot \mathbf{a}(\theta)} \quad (9)$$

C. Multiple Signal Classification

Multiple Signal Classification (MUSIC) algorithm is a super resolution parametric estimator. It is a very high

resolution estimator with less computational complexity compared to other super resolution algorithms. MUSIC will be efficient when the number of antenna elements, K is greater than the number of signal sources M that in turn, must be uncorrelated. The algorithm steps is as follows

- Calculate eigenvalues and eigenvectors for the matrix R which will be $K \times K$.
- M and the noise variance σ^2 can be determined knowing that the minimum eigen value equals σ^2 and repeated for $K - M$ times.
- Group the noise subspace eigenvectors in a set $[e_{M+1}, \dots, e_K]$
- find $P(\theta) = \frac{\mathbf{a}^H \cdot \mathbf{a}}{\sum_{i=M+1}^K |\mathbf{a} \cdot \mathbf{e}_i|^2}$
- At every DoA, P will peak to ∞

In case of correlated signals, which is typical in multipath scenarios, especially in indoor environment, MUSIC will fail in calculating the DoA. To solve this problem, MUSIC should be combined with a spacial smoothing technique to restore its capabilities. MUSIC accuracy also increases with high SNR and when K increases and the number of snapshots increase.

D. Estimation of Signal Parameters via Rotational Invariance Technique (ESPRIT)

ESPRIT is another super resolution parametric technique. It is like the MUSIC technique but has the advantage of eliminating the array calibration required by MUSIC [7]. The algorithm can be implemented as in the following steps:

- Calculate the matrix R from N samples of the signal according to

$$R_{\text{ESPRIT}} = \mathcal{E}(\mathbf{r}_{\text{ESPRIT}}(t) \cdot \mathbf{r}_{\text{ESPRIT}}^H(t)) \quad (10)$$

$$\mathbf{r}_{\text{ESPRIT}}(t) = [\mathbf{r}_L(t) \cdot \mathbf{r}_U(t)]^T \quad (11)$$

$$\mathbf{r}_L(t) = [\mathbf{r}_1(t) \dots \dots \mathbf{r}_{K-1}(t)] \quad (12)$$

$$\mathbf{r}_U(t) = [\mathbf{r}_2(t) \dots \dots \mathbf{r}_K(t)] \quad (13)$$

- Calculate the generalized eigenvalue problem $R_{\text{ESPRIT}} \mathbf{E} = \Sigma \mathbf{E} \Lambda$ (14)

where

$$\Sigma = \begin{pmatrix} \mathbf{I} & \mathbf{Q}_{-1} \\ \mathbf{Q}_{+1} & \mathbf{I} \end{pmatrix} \quad (15)$$

Λ is a matrix whose diagonal holds the eigenvalues λ_i , which are ordered descendingly, $i = 1 \dots 2K$, $\mathbf{E}$ holds the eigenvectors, $\mathbf{I}$ is a $K \times K$ identity matrix, and $\mathbf{Q}_{-1}$ and $\mathbf{Q}_{+1}$ are $K \times K$ zeros matrices with unity elements under and upper their diagonal respectively.

- The number of sources M can be estimated from the number of eigenvalues exceeding the noise floor.
- Compose the subspace matrix $\mathbf{E}_{\text{sub}}$ from $\mathbf{E}$ by selecting the M eigenvectors corresponding to the eigenvalues that exceed the noise floor.

- Obtain E_{subL} and E_{subU} from E_{sub} in a way that resembles decomposition in previous steps.

- Compute the matrix F according to

$$F = [E_{subL}^H, E_{subU}^H]^T \cdot [E_{subL}, E_{subU}]$$

- Find eigenvalues Λ_F and eigenvectors E_F of F and rearrange them descendingly.

- Arrange E_F into four equidimensional submatrices according to

$$E_F = \begin{pmatrix} E_{F1} & E_{F2} \\ E_{F3} & E_{F4} \end{pmatrix}$$

- Calculate the eigenvalues Λ_Ω of the rotation operator Ω given by

$$\Omega = -\frac{E_{F2}}{E_{F4}}$$

- Obtain the DoA vector τ as

$$\tau = \sin^{-1}\left(\frac{c}{\omega_c d} \arg(\Lambda_\Omega)\right)$$

III. MIMO CHANNEL MODELLING

Many channel models were developed for testing and simulating wireless channels. The main purpose for many models is to measure the channel capacity [9] or to deal with the bit error rate (BER) performance [10]. The channel models can be classified into: stochastic and deterministic models [11]. Stochastic models, further, include geometrical models, parametric models, and correlation based models. Deterministic models include recorded impulse response and ray tracing model. More details about channel models that are standardized by the IEEE can be found in [12].

It is evident that, for examining or simulating positioning techniques, a full knowledge of the wireless channel is needed. It is known that as the frequency of operation increases, the signal tends to behave like an optical signal that can be reflected, refracted or diffracted. This suggests the modelling of the MIMO wireless channel using the "ray tracing" model. This model is accurate for site specific modelling. However, it is complex and computationally heavy. In this paper, a simplified method for calculating ray traces is introduced. This facilitates the modelling and simulation of the practical channels for the indoor single-antenna and the multi-antenna systems.

A. Ray Tracing

In high-frequency bands (WLAN frequency bands are among them), the electromagnetic waves can be treated as light rays, making it possible to predict the wave paths with an acceptable accuracy. The ray tracing model (RTM) is a technique used to predict all the paths taken by the electromagnetic waves from a transmitter to a reception point. Through the paths, some phenomena that influence the received signal may occur. The most significant of these phenomena are reflections, refractions and

diffractions [13]. However the resulting channel response depends on site dimensions and antenna locations, it provides detailed information about multipath delays, direction of arrival, direction of departure and path loss.

IV. CHANNEL IMPLEMENTATION VIA RAY TRACING

The implementation method employed in this paper is through using the vector analysis and the numerical iterations. Consider the transmitting antenna is located at point P_t and the receiving antenna is located at point P_r while the reflection point on the i^{th} wall is $P_{i,ref}$.

The length of the direct path is

$$L_d = \|P_r - P_t\| \quad (16)$$

The unit vector, $\vec{u}_d$ calculated as

$$\vec{u}_d = \frac{P_r - P_t}{L_d} \quad (17)$$

This unit vector is useful in finding the direction of arrival (DoA) or direction of departure (DoD)

To find the single interaction point $P_{i,ref}$, a search for that point is performed along the i^{th} wall so that

$$\vec{u}_{r-ref} = -\vec{u}_{t-ref} \quad (18)$$

Where

$$\vec{u}_{r-ref} = \frac{P_{i,ref} - P_r}{\|P_{i,ref} - P_r\|} \quad (19)$$

And

$$\vec{u}_{t-ref} = \frac{P_{i,ref} - P_t}{\|P_{i,ref} - P_t\|} \quad (20)$$

For double interaction with the i^{th} wall and j^{th} wall, two points $P_{i,ref}$ and $P_{j,ref}$ is to be found to obey the following relations

$$\begin{aligned} \vec{u}_{t-ref} &= -\vec{u}_{ij-ref} \\ \vec{u}_{r-ref} &= -\vec{u}_{ij-ref} \end{aligned} \quad (21)$$

After finding interaction points and unit vectors to the receiving antenna and transmitting antenna, the DoA can be calculated for a multi-antenna receiver or transmitter if the line passing through the antenna elements (considering a Uniform Linear Array (ULA) as in most IEEE802.11n Access Points (APs)) makes an angle θ_a with horizontal.

$$\vec{u}_{ULA} = \cos\theta_a \vec{a}_x + \sin\theta_a \vec{a}_y \quad (22)$$

Where $\vec{a}_x$ and $\vec{a}_y$ are the unit vectors in x and y directions respectively.

$$\begin{aligned} DoA &= \cos^{-1} u_{ULA,r} \odot u_{r-ref} \\ DoD &= \cos^{-1} u_{ULA,t} \odot u_{t-ref} \end{aligned} \quad (23)$$

Where $u_{ULA,t}$ designates the unit vector for transmitting array with angle $\theta_{a,t}$ and $u_{ULA,r}$ is the unit

vector of the receiving array with angle $\theta_{a,r}$ and they are found according to above equations.

In this way, DoAs and DoDs can be calculated for each ray leaving or reaching an antenna element. This introduces a precise knowledge of them that is required when testing DoA algorithms in simulation.

The total path length from an antenna element in the transmitter to another at the receiver can be calculated from the points of interactions and positions of antenna elements. The path delay τ_k is then as $\tau_k = \frac{L_k}{c}$, where L_k is the length of the k^{th} path.

V. EVALUATION SYSTEM SETUP

To achieve the performance comparison between different DoA estimation algorithms in indoor environment, the MIMO-OFDM system shown in Fig. 2 was established. In this model, the estimation is performed using the baseband signals. The baseband signals are generated via the “MIMO-OFDM encoding” block that employs an orthogonal space time block coding (OSTBC) with the signal structure as specified in the IEEE 802.11n [14]. The output signals then modulate a carrier in the “Modulation” block to generate the signal that excites the wireless channel. The receiver demodulates the received signal in the “Demodulation” block. The received baseband signal is then processed in the “DoA-Estimation-Algorithm” block for DoA estimation. The baseband signal may also be processed via the “Channel Equalization” and the “MIMO-OFDM-Decoder” blocks for obtaining the transmitted data. The receive- and transmit-antenna elements are omni-directional monopoles. The MIMO configurations that used are 2×4 and 3×4 , where the first number indicates the number of transmit antenna elements while the second number indicates the number of receive antenna elements. These configurations are focused as it is assumed that the DoA estimation algorithm will be implemented in the WLAN’s access point that analyses the received signal from a mobile unit. The mobile unit may transmit at lower data rate than the access point.

The channel model employed is the ray-tracing model [5], [15] discussed above. Line of sight paths (LOS), single

reflections and double reflections are the only components that are simulated done in two-dimensions (2D) to reduce computational load. Higher order reflections introduce negligible weak signals.

VI. RESULTS AND DISCUSSIONS

A. Ray Tracing Channel Model

Visualization of the channel simulated via RTM is shown in Fig. 3. This figure shows the walls bounding a 100×200 meters place and the electromagnetic waves propagate and reflect until they reach to the antenna elements at the receiver. The configuration simulated is a 3×4 MIMO. The channel impulse response (CIR) for that channel is shown in Fig. 4. The fading caused by multipath is clear, which proves the usefulness and the ability of fully modelling the MIMO channels with RTM.

The relative error e_{DOA} in DoA estimation is calculated as

$$e_{DOA} = |\hat{\theta}_{DOA} - \theta_{DOA}|$$

where $\hat{\theta}_{DOA}$ is the estimated DoA and θ_{DOA} is the true DoA. The noise is modelled by AWGN process. Spatial smoothing is applied with MUSIC and ESPRIT to increase their efficiency because the signals in multipath channel are always correlated.

Fig. 5 introduces the measured error e_{DOA} versus the d/λ ratio, (d is the inter-element distance of the receiving antenna array and λ is the wavelength of the signal) for the 2×4 MIMO. Dark blue areas indicate estimation error less than 5 degrees. As these areas become contiguous, a continuous range of incidence angles can be estimated at certain d/λ ratio. It is preferable to select d/λ that maximizes the range for successful and accurate estimation of incidence angles. This results in a least attainable estimation error. It is obvious that, as the number of receiving elements increases, the estimation error decreases. However, the number of elements is always limited in known commercial access points. Hence, to increase estimation accuracy, number of elements should be increased or other antenna types shall be used. Table I is deduced directly from Fig.5. It summarizes the maximum attainable range of incidence angles for the

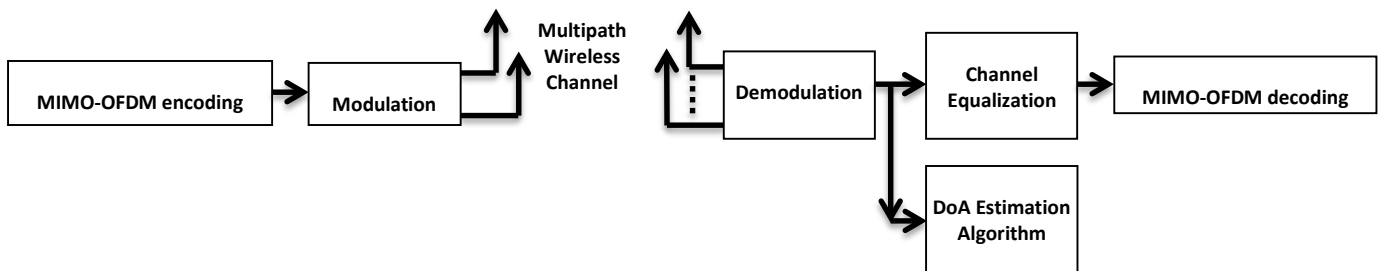


FIG. 2: MIMO-OFDM SYSTEM MODEL USED FOR DOA ESTIMATION

least estimation errors. It shows that the ESPRIT and the MUSIC techniques lead to the best accurate estimation range for both configurations. In ESPRIT, the antenna elements should be separated by 0.549λ to achieve a maximum range for accurate estimation of 105.205° for the indoor environment, while MUSIC should operate also at 0.529λ to obtain an estimation range of 100.85° . The results for BS and MVDR techniques proved that they are unsuitable for DoA estimation in indoor environments.

TABLE I: MAX ATTAINABLE RANGE OF INCIDENCE ANGLES FOR INDOOR ENVIRONMENT

	Max. detectable, angle	Min. detectable, angle	Max. range	Antenna element spacing related to λ	Antenna element spacing (cm) at 2.4 GHz
Beam Scan	10.549	-31.993	42.543	0.173λ	2.16
MVDR	14.544	-40.845	55.389	0.167λ	2.09
MUSIC	51.992	-48.857	100.850	0.529λ	6.6
ESPRIT	51.818	-53.386	105.205	0.549λ	6.86

Fig. 6 shows how these minimum obtained estimation error for different estimation techniques and with the two MIMO configurations, 2×4 and 3×4 with inter-element spacing of 0.5λ vs. signal to noise ratio (SNR). It is clear that while the increase in number of transmitting antennas the estimation performance is negatively affected as the increased number of signals introduce extra interferers which acts like increasing the noise level. This decrease in estimation performance is around 2° on average, which is not significant on small ranges as in indoor environments. It also shows that after 10 dB , the MUSIC and the ESPRIT performance is rapidly improved until the SNR reach 20 dB . After this SNR, e_{DoA} starts to decrease below 20° for the MUSIC and below 15° for the ESPRIT. The e_{DoA} starts to decrease below about 9° starting from approximately 30 dB . It reaches 5° approximately at 40 dB . From 20 dB to 30 dB , ESPRIT performs better than MUSIC. Totally, ESPRIT combined with the 2×4 MIMO configuration performs better than all other techniques at all the values of SNR.

VII. CONCLUSIONS

Location estimation using the direction of arrival (DoA) detection is a promising technique for the IEEE 802.11n, ac, as they introduced the MIMO for the air interface. MIMO-OFDM coding with OSTBC is used for DoA estimation through the IEEE802.11n WLAN signals in indoor environments. DoA estimation algorithms such as beam scan (BS), minimum variance distortion-less response (MVDR), multiple signal classification (MUSIC), and estimation of signal parameters via rotational invariance technique (ESPRIT) are compared regarding the range of accurate and contiguous estimation of DoA and the effect of the inter-element spacing on the estimation accuracy at the receiving antenna. The MIMO Wireless

They can cover only ranges of 42.543° at element spacing of 0.173λ (2.16 cm at 2.4 GHz) for the BS and 55.389° at element spacing of 0.167λ (2.09 cm at 2.4 GHz) for the MVDR. At these small inter-element spacing, coupling between the electromagnetic fields for the individual elements may reduce the capacity of the MIMO channel. This is clear as shown in Table I. The inter-element spacing of 6.6cm 6.86cm at 2.4 GHz for the MUSIC and the ESPRIT will lead to negligible inter-element coupling.

Channel was simulated via the ray-tracing model (RTM) for gaining all the details of the individual signal rays comprising the received multipath signals. It is found that the increase in the number of antenna elements at the transmitter decreases the performance of the estimation techniques as it introduces more interferers, which deteriorate the estimation accuracy. However, MUSIC and ESPRIT techniques proved their accuracy for DoA estimation in indoor environments. It is noted that they can support a wide range of incidence-angles, compared to BS and MVDR, while keeping the maximum accuracy. It is also proved that the antenna elements at the receivers should be spaced by approximately 0.5λ for maximum contiguous measurable DoA angle range if MUSIC or ESPRIT is used for DoA estimation. This is important to keep the system capacity from reduction by the inter-element coupling. Thus, MUSIC and ESPRIT is the most suitable techniques for DoA estimation through WLANs in indoor environments.

REFERENCES

- [1] C. M. Takenga and K. Kyamakya, "Robust positioning system based on fingerprint approach," in Proceedings of the 5th ACM international workshop on Mobility management and wireless access, New York, NY, USA, 2007, pp. 1-8.
- [2] T. J. S. Khazada, A. R. Ali, and A. S. Omar, "Time Difference of Arrival estimation using super resolution algorithms to minimize Distance Measurement Error for indoor positioning systems," in Multitopic Conference, 2008. INMIC 2008. IEEE International, 2008, pp. 443-447.
- [3] D. Munoz, F. B. Lara, C. Vargas, and R. Enriquez-Caldera, Position Location Techniques and Applications: Academic Press, 2009.
- [4] M. K. Ragheb, S. H. Elramly, and S. Mostafa, "The effect of varying the input parameters on the performance capabilities of three different DOA estimation algorithms using smart antennas," in Radio Science Conference, 2003. NRSC 2003. Proceedings of the Twentieth National, 2003, pp. B6 - 1-9.
- [5] Z.-Y. Liu, L.-X. Guo, and T.-Q. Fan, "Microcellular propagation prediction model based on an improved ray tracing algorithm," J. Opt. Soc. Am. A 30, 2372-2380 (2013)

- [6] M. H. C. Dias and G. L. Siqueira, "Indoor 1.8 GHz AOA-TDOA Measurements with Simple Channel Sounding Approaches," *Journal of Microwave and Optoelectronics*, vol. Vol. 5, December 2006.
- [7] A. Bensky, *Wireless Positioning Technologies and Applications*. Norwood, MA, USA: Artech House, Inc., 2007.
- [8] P. Rong and M. L. Sichitiu, "Angle of Arrival Localization for Wireless Sensor Networks," in *Sensor and Ad Hoc Communications and Networks*, 2006. SECON '06. 2006 3rd Annual IEEE Communications Society on, 2006, pp. 374-382.
- [9] J. W. Wallace and M. A. Jensen, "Modeling the indoor MIMO wireless channel," *Antennas and Propagation, IEEE Transactions on*, vol. 50, pp. 591-599, 2002.
- [10] K. Yoon Hyun and K. Jin Young, "Performance comparison of 802.11n system with WLAN channel model," in *Communications and Information Technology, 2009. ISCIT 2009. 9th International Symposium on*, 2009, pp. 790-795.
- [11] R. Hoppe, J. Ramuh, H. Buddendick, O. Stabler, and G. Wolfle, "Comparison of MIMO Channel Characteristics Computed by 3D Ray Tracing and Statistical Models," in *Antennas and Propagation, 2007. EuCAP 2007. The Second European Conference on*, 2007, pp. 1-5.
- [12] (2004, 802.11n Channel Models. Available: <https://mentor.ieee.org/802.11/dcn/03/11-03-0940-04-000n-tgn-channel-models.doc>
- [13] S. Grubisic, W. P. Carpes, and J. P. A. Bastos, "Optimization Model for Antenna Positioning in Indoor Environments Using 2-D Ray-Tracing Technique Associated to a Real-Coded Genetic Algorithm," *Magnetics, IEEE Transactions on*, vol. 45, pp. 1626-1629, 2009.
- [14] "IEEE Standard for Information technology--Telecommunications and information exchange between systems--Local and metropolitan area networks--Specific requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 5: Enhancements for Higher Throughput," *IEEE Std 802.11n-2009 (Amendment to IEEE Std 802.11-2007 as amended by IEEE Std 802.11k-2008, IEEE Std 802.11r-2008, IEEE Std 802.11y-2008, and IEEE Std 802.11w-2009)*, pp. c1 -502, 29 2009.
- [15] P. Almers, E. Bonek, A. Burr, N. Czink, M. Debbah, V. Degli-Esposti, H. Hofstetter, s. Ky\\\"o, P., D. Laurenson, G. Matz, and others, "Survey of channel and radio propagation models for wireless MIMO systems," *EURASIP Journal on Wireless Communications and Networking*, vol. 2007, pp. 56-56, 2007.

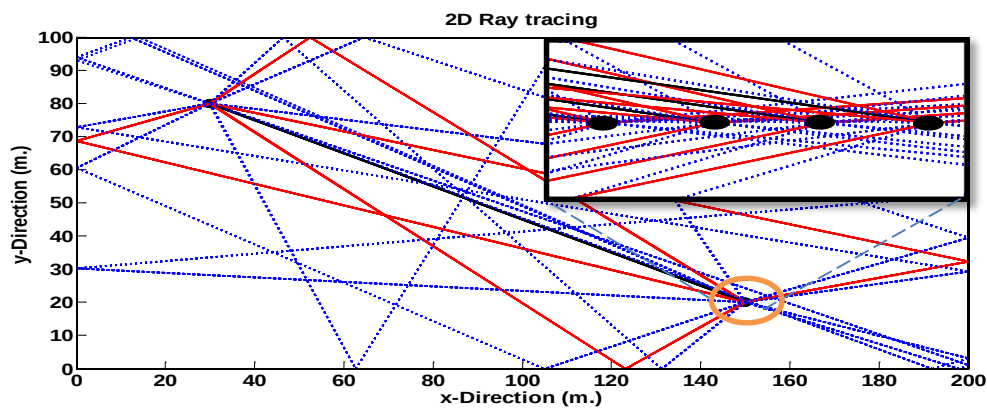


FIG. 3: RAYS THAT TRANSMITTED AND COULD BE RECEIVED BY THE ANTENNA AT THE RECEIVER USING RAY-TRACING MODEL

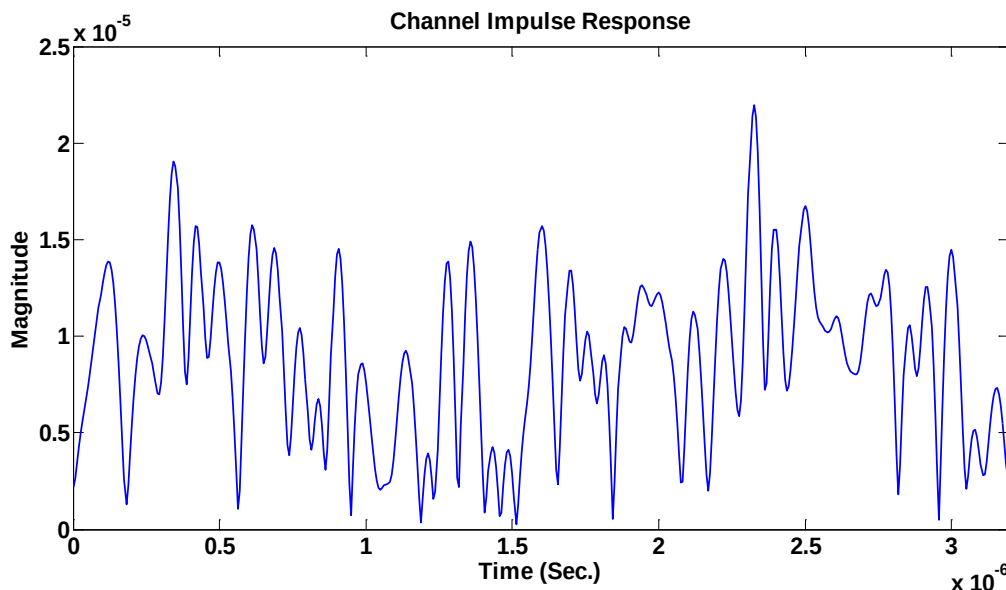


FIG. 4: CHANNEL IMPULSE RESPONSE (CIR) OF THE CHANNEL SIMULATED.

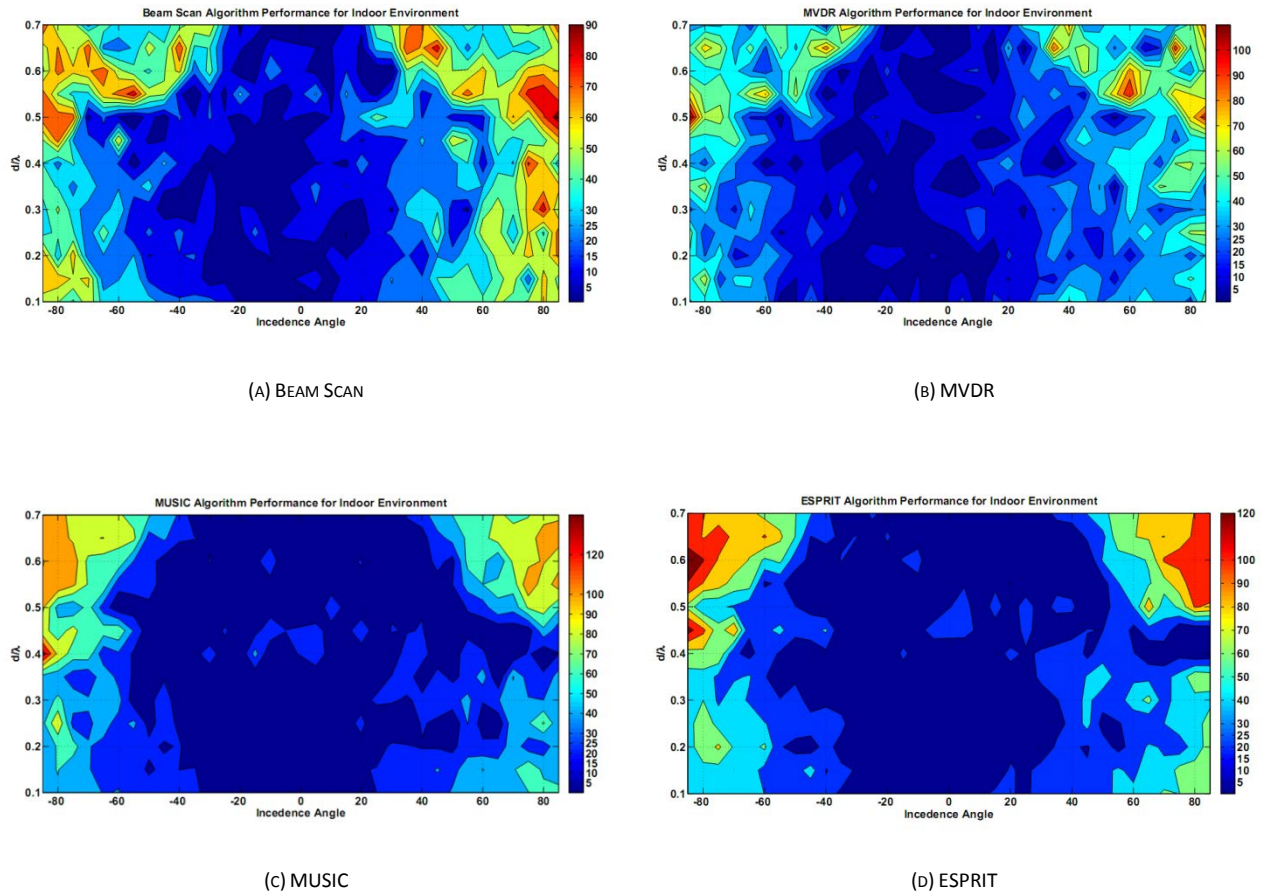


FIG. 5: DOA ESTIMATION ERROR VS. d/λ VS. INCIDENCE ANGLE FOR INDOOR ENVIRONMENT

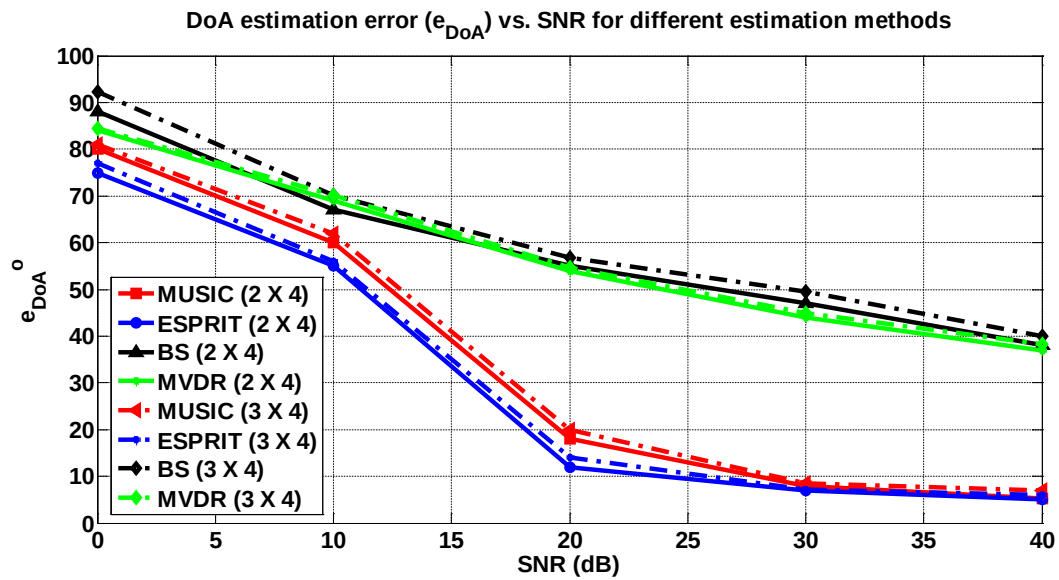


FIG. 6: DOA ESTIMATION ERROR VS. SNR